The memory squeeze

With time, technology gets faster, smaller and cheaper. This is what helps with the trickle down effect of technology as newer and better features slowly become democratised. Of late, things aren't getting "cheaper" by any sense of the word. For years, memory was the quiet enabler of better budget technology. It was rarely the part that got top billing, but it was often the reason affordable products improved so quickly. A cheap smartphone could suddenly offer more RAM. A budget laptop could move from a cramped SSD to something usable. A small maker of mini PCs, routers, cameras, handheld consoles or smart home devices could build a decent product around commodity components and still hit a price that did not scare away buyers.

The current memory crunch is usually discussed through the lens of large companies. Even Apple, a company known for freezing prices for 3-5 years by buying up all manufacturing capacity of the latest silicon, had to raise pricing. And it's not just Apple. Microsoft's AI PC ambitions may run into BOM (Bill of Materials) pressure. Console makers, laptop vendors and smartphone brands have already begun to rethink storage tiers. Those are alarming movements, but they are not the most worrying part. The bigger issue is what happens to the small manufacturers that live and die in the lower half of the market.

A large company has options. It can lock in supply through long-term contracts. It can spread higher component costs across a wider product portfolio. It can negotiate harder, delay upgrades, use financing schemes, push buyers towards pricier models, or absorb a bad quarter if needed. A fledgling hardware company does not have those cushions. It has a thinner margin, a smaller purchase order, less bargaining power, and a customer who is far more price-sensitive. This is where the memory crunch becomes more than a supply chain inconvenience. It becomes a market filter.

When DRAM and NAND prices rise, the immediate response for a budget product maker is not always to raise prices. In many cases, that simply does not work. A Rs 12,000 phone, a Rs 25,000 laptop, a low-cost security camera, a compact NAS, an Android TV box, or a student-focused tablet exists because it sits at a psychological price point. Add 10 or 15 percent to the final price and the product may no longer make sense to its intended buyer. The big brands can sometimes justify that increase with marketing muscle. A smaller brand often cannot.

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Managing Editor

So the compromise starts elsewhere. Reduce RAM. Cut storage. Move to older memory standards. Use slower chips. Reuse an older board design. Stay on older process nodes because the newer ones are either too expensive or unavailable in small volumes. On paper, this keeps the product alive. In practice, it slowly hollows it out. The problem is that older process nodes are not an infinite safety net. As fabs chase higher-margin opportunities, older capacity can be repurposed, consolidated or phased out. Mature nodes will continue to exist, especially for automotive, industrial and embedded products, but not every old part stays cheap or widely available forever. Once the industry starts prioritising high-bandwidth memory, advanced packaging and data-centre-grade components, the long tail of low-margin consumer parts becomes easier to neglect.

That creates a trap for smaller manufacturers. They can't move up to newer components because prices are high and supply is tight. They can't rely on older components because supply chains are tougher to maintain. They cannot charge more because the customers came to them because they were cheaper. The result will not be dramatic at first. It will look like fewer variants, longer refresh cycles, delayed launches and conservative specs. A product that should have 8 GB RAM may ship with 4. A device that should have moved to faster storage may stick with eMMC. A mini PC maker may keep an ageing platform alive for longer. A smart device company may cancel its cheapest model because the economics no longer add up.

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